

EMPOWERING EDUCATION THROUGH AI-SUPPORTED E-LEARNING

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Higher education institutions globally struggle with student retention and engagement due to rigid, "one-size-fits-all" pedagogical models that fail to address diverse learning paces and backgrounds. While AI presents transformative opportunities, current educational platforms remain highly fragmented, typically isolating content delivery, static tutoring, and assessment without providing cohesive, dynamic personalization. This creates a critical research gap: the absence of an integrated system that truly aligns instructional design with an adaptive, self-paced learning experience. To bridge this gap, this project proposes an AI-Powered Personalized Course Creation and Discovery Platform. The solution is an integrated, multimodal suite comprising four core AI modules coordinated by a Central Knowledge Orchestrator. It features an interactive infographic generator, a bidirectional audio review component, an adaptive assessment engine using continuous knowledge tracing, and a multilingual handwritten note extractor supporting English, Tamil, and Sinhala. The system leverages open-source Large Language Models (LLMs), natural language processing (NLP) to accommodate mixed-language queries like Sinhlish and Thanglish, and machine learning frameworks such as TensorFlow and PyTorch. The expected outcome is a unified dashboard delivering personalized multimedia lessons, dynamic flashcards, and adaptive quizzes that update a real-time learner model to close knowledge gaps. Socially, this platform democratizes quality education, mitigates equity issues, and fosters self-directed learning, directly advancing SDG 4. Commercially, it targets the EdTech and Higher Education sectors, promising to decrease time-to-competency for graduates and improve institutional retention and reputation.

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LIST OF ABBREVIATIONS

- AI: Artificial Intelligence
- LLM: Large Language Model
- NLP: Natural Language Processing
- ML: Machine Learning
- OCR: Optical Character Recognition
- ASR: Automatic Speech Recognition
- TTS: Text-to-Speech
- XAI: Explainable Artificial Intelligence
- JSON: JavaScript Object Notation
- API: Application Programming Interface
- UI/UX: User Interface / User Experience
- LMS: Learning Management System
- SaaS: Software as a Service
- GPU: Graphics Processing Unit
- OOP: Object-Oriented Programming
- WBS: Work Breakdown Structure
- DSR: Design Science Research
- SDG: Sustainable Development Goal
- IRB: Institutional Review Board
- STEM: Science, Technology, Engineering, and Mathematics

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1. INTRODUCTION

1.1 Background and Context

Higher education institutions globally face persistent challenges in student retention, engagement, and academic progression, which are exacerbated by diverse student populations with varying learning needs, paces, and backgrounds. The traditional "one-size-fits-all" pedagogical model struggles to address this diversity, often leading to disengagement, knowledge gaps, and suboptimal learning outcomes. The primary users affected by these issues are university students, particularly undergraduates, and educators. Students from non-traditional backgrounds or those with different learning styles are disproportionately impacted by rigid curricula, which contributes to equity issues in education as uniform teaching can widen the achievement gap. Furthermore, this rigid approach fails to foster crucial self-directed learning skills and deep engagement.

While technology-enhanced learning and Artificial Intelligence (AI) present transformative opportunities, current educational solutions are highly fragmented. Many platforms focus either on simple content delivery, static tutoring, or automated assessment without integrating true personalization and intelligent content discovery. Specifically, in the domain of data synthesis and study material generation, current tools—including Google NotebookLM—typically generate static or linear visual summaries from user notes. This highlights a critical technical gap: the lack of context-aware interactivity and user-driven narrative control that allows learners to actively explore and interrogate their study materials.

To address this specific gap, this research proposes the development of an intelligent, dynamic infographic generator that creates interactive and data-rich visual narratives from structured notes or research data. The system utilizes a schema driven by an open-source Large Language Model (LLM) to extract and structure key entities, relationships, and statistics. By implementing multiple visualization templates (such as hierarchical, timeline, and network views) using open-source libraries, the tool transforms static charts into explorable data stories. An interactive interface will allow users to dynamically filter data, click on chart elements to drill down into specific details or source quotes, and toggle between visualization types. To further enhance collaborative learning, the component integrates a rule-based AI model to suggest suitable chart types and enables real-time collaboration so multiple users can view, annotate, and discuss the same infographic simultaneously.

This project directly advances Sustainable Development Goal (SDG) 4 (Quality Education) by aiming to make inclusive, equitable, and effective learning a reality through personalized adaptation. By leveraging technology to enhance educational delivery, it specifically promotes SDG 4.c. Academically, this problem aligns perfectly with the Centre of Excellence in AI (CoEAI) research cluster, requiring deep research in core AI domains such as natural language processing for content extraction and data analytics for visual insight generation.

1.2 Literature Review

2.1 Existing Research Approaches

The integration of Artificial Intelligence (AI) into educational frameworks has been a primary focus of recent academic research, driven by the need to overcome the limitations of traditional, uniform instruction. Storytelling and visualization have long been integral to human communication and learning, but recent advances in natural language processing (NLP) and machine learning (ML) have revolutionized how educational content is generated and adapted. The emergence of generative AI introduces both significant opportunities and notable challenges, particularly in fields like STEM education where precision and critical thinking are paramount (Ru & Li, 2025).

Recent studies emphasize that AI-driven adaptive learning platforms and intelligent tutoring systems (ITS) significantly outperform traditional instruction by tailoring educational pathways to individual needs. Researchers have extensively explored how technologies like Deep Reinforcement Learning and NLP enable these systems to dynamically adjust content, thereby fostering deeper engagement and self-directed learning. For instance, studies by Al Balushi and Al Harthi (2024), as well as Dutta et al. (2024), highlight the critical role of adaptive learning pathways in advancing personalization within online education ecosystems. Furthermore, Fang and Xie (2025) demonstrate that the use of intelligent online learning tools significantly impacts college students' self-directed learning abilities. The very nature of how learners interact with digital content is also under scrutiny, with research suggesting that interactive elements, such as those found in infographics, can have measurable effects on user affect, including anxiety levels (Maria & Huang, 2024).

A growing area of interest is the transparency and interactivity of these AI systems. Majumdar et al. (2023) explored the use of Explainable AI (XAI) recommendations in schools, extracting patterns from learning logs to better support self-directed study. Additionally, the academic community is closely examining the effectiveness of specific Large Language Models (LLMs). Teo and Chow (2025) conducted exploratory studies on student perspectives regarding AI-powered tools like Deepseek, ChatGPT, and Gemini, while Rozputnia et al. (2025) analyzed the future trends of integrating Google Workspace and Gemini into higher education. Notably, Banihashemi et al. (2025) conducted controlled experiments comparing human-generated content against AI systems like NotebookLM, revealing that AI-generated educational materials can rival human-created resources in quality and engagement. Concurrently, technical advancements in AI are pushing the boundaries of content creation, with research into LLM-powered storytelling frameworks (Pantola et al., 2025) and multi-modal generation models for creating images (Wang et al., 2025; Zhou & Wang, 2025), paving the way for richer, more automated educational content.

2.2 Existing Products and Services The current commercial landscape features diverse AI-driven tools emphasizing personalization, often integrating large language models (LLMs) for content generation and machine learning for adaptation. Google's NotebookLM targets researchers and students by processing uploaded sources to deliver tailored insights, explanations, and audio overviews. Similarly, systems like Learn Your Way transform static PDFs into dynamic courses using a blend of AI and pedagogy, while Tutor AI creates custom courses and adapts paths for lifelong learners.

Other platforms focus heavily on tutoring and assessment. ChatGPT's Study Mode adjusts lessons based on user levels, offering quizzes and flashcards. Claude for Education provides Socratic guidance specifically designed for higher-education users, and Khanmigo offers

standards-aligned tutoring integrated with vast educational content. Tools designed for rapid study material generation, such as StudyX , Gizmo , Mindgrasp , and Diffit, focus on extracting information to generate notes, spaced-repetition flashcards, and quizzes.

2.3 Limitations of Existing Systems Despite these technological advancements, significant limitations and research gaps remain. The most prominent structural limitation is fragmentation; many platforms focus on either simple content delivery, static tutoring, or automated assessment without integrating true personalization and dynamic pathway adaptation into a cohesive ecosystem. Existing tools often address only one side of the equation—either content creation or learner tutoring—leading to a disconnect between instructional design and the adaptive learning experience.

Specific to data synthesis and visual learning, current tools, including Google NotebookLM, typically generate static or linear visual summaries from user notes. Their primary limitation is a lack of context-aware interactivity and user-driven narrative control. They fail to transform static charts into explorable data stories where users can interrogate the visualization by filtering, drilling down, or switching perspectives to uncover insights. While foundational research has established the benefits of interactive multimedia modules (Sanderson et al., 1996) and visualization techniques for understanding document collections (Spangler, Kreulen & Lessler, 2002), modern implementations have not yet fully realized this interactive potential. Furthermore, techniques for interactive image segmentation (Chen et al., 2010) have existed for over a decade, yet they have not been effectively adapted for educational data visualization to allow for deep, user-driven exploration of content.

Furthermore, the widespread adoption of these systems faces several ethical and technical hurdles. Privacy and security risks arise from extensive data collection, while algorithmic bias in training data can disadvantage marginalized groups, exacerbating educational inequities. There is also a distinct over-reliance on proprietary, "black box" AI architectures, which lack the transparency and modifiability offered by open-source models. Finally, an over-reliance on AI may erode critical thinking, underscoring the need for platforms that actively engage the user through interactive, explorable learning assets rather than passively feeding them generated content.

1.3 Research Gap

Significant advancements have been made in educational technology and AI-driven platforms, a profound disconnect remains between instructional design and the adaptive learning experience. Existing tools often address only one side of this equation—either simple content delivery or automated assessment—resulting in highly fragmented solutions. In the specific domain of data synthesis and study material visualization, current approaches are fundamentally insufficient because they rely on passive consumption. Advanced tools currently on the market, including Google NotebookLM, typically generate only static or linear visual summaries from user notes. What is critically missing is context-aware interactivity and user-driven narrative control. These existing systems function as rigid frameworks that do not allow learners to deeply interrogate the visualizations, drill down into specific data points, or dynamically switch perspectives to uncover hidden insights. This limitation prevents the realization of a fully responsive educational environment and fails to foster the crucial self-directed learning skills required in higher education.

To address these shortcomings, the novelty of the proposed interactive infographic generator lies in its ability to transform static charts into explorable, data-rich stories. Unlike existing platforms, this system utilizes an open-source Large Language Model (LLM) to intelligently extract and structure key entities and relationships, building upon emerging frameworks for AI-powered storytelling (Pantola et al., 2025). This is paired with a lightweight AI model that automatically suggests the most suitable visualization template (e.g., hierarchical, timeline, or network) based on the analyzed data. The creativity of the solution is further demonstrated through a dynamic interface that allows users to seamlessly filter data and click on specific chart elements to instantly view the original source quotes—an application of interactive principles advocated in early visualization research (Spangler, Kreulen & Lessler, 2002) and enabled by modern interaction techniques (Chen et al., 2010). Furthermore, by incorporating real-time collaborative features—allowing multiple users to simultaneously view, annotate, and discuss the same infographic—and utilizing a transparent, open-source architecture, this component creates a highly competitive, evidence-based alternative to existing proprietary "black box" systems.

2. OBJECTIVE

2.1 Main Objective

To develop an intelligent, dynamic infographic generator utilizing open-source Large Language Models (LLMs) to transform structured educational notes and research data into interactive, explorable visual narratives, thereby enhancing visual knowledge exploration and user-driven narrative control compared to existing static visualization tools.

2.2 Specific Objectives

- To design an AI-driven schema utilizing open-source Large Language Models (LLMs) for accurately extracting and structuring key entities, relationships, and statistical data from unstructured user notes.
- To implement a dynamic visualization engine that incorporates multiple open-source templates (such as hierarchical, timeline, and network views) and integrates a lightweight AI model to automatically suggest the most appropriate chart type based on the analyzed data structure.
- To develop a context-aware, interactive user interface that empowers learners to actively filter data, toggle between visual perspectives, and click specific chart elements to seamlessly drill down into original source quotes and detailed insights.
- To engineer real-time collaboration capabilities within the visual environment, enabling multiple users to simultaneously view, annotate, and discuss the generated data stories, thereby fostering peer-to-peer engagement.
- To construct an export module that allows users to seamlessly publish their customized visualizations as interactive HTML files or high-quality static images for inclusion in external academic reports.

3. METHODOLOGY

3.1 Research Approach

The methodology for developing the interactive infographic generator follows a Design Science Research (DSR) approach, as it focuses on the creation, development, and evaluation of an innovative technological artifact intended to solve a specific educational challenge. This is complemented by an experimental approach during the validation phase, where the proposed interactive visualization models will be tested against traditional static models to measure improvements in user engagement and data comprehension.

3.2 Technical Methods and Algorithms

The core processing pipeline relies heavily on Natural Language Processing (NLP) techniques. The system will implement a custom extraction schema powered by an open-source Large Language Model (LLM) to accurately parse unstructured study notes and extract key entities, relationships, and statistical data. Once the data is structured, the system will utilize a rule-based or lightweight machine learning algorithm to analyze the data's dimensional structure and automatically suggest the most suitable visualization template (e.g., hierarchical, timeline, or network). The rendering engine will then construct the explorable charts, embedding metadata to allow users to click elements and retrieve source quotes.

3.3 Tools and Platforms

- **Core Programming:** Python will serve as the primary foundational language, utilizing its robust data structures and Object-Oriented Programming (OOP) capabilities.
- **AI and NLP:** Open-source models and NLP libraries, such as those provided by Hugging Face, spaCy, and NLTK, will be utilized for content extraction and analysis. Utilizing open-source LLMs ensures transparency and avoids the limitations of proprietary "black box" systems.
- **Visualization:** Open-source front-end visualization libraries will be employed to construct the dynamic templates and facilitate the real-time collaborative annotation features.

3.4 Data Collection and Ethical Considerations Given strict privacy constraints and the lack of high-quality localized educational datasets, the project adopts a triangulated hybrid data strategy. The primary source for initial training and logic building will be LLM-generated synthetic data (using models like Gemini 3 Pro or Deepseek via Knowledge Distillation) to create structured course content and simulated interactions. Public benchmarking datasets, such as "EdNet" and "ASSISTments," will also be used.

Ethical clearance from the Institutional Review Board (IRB) is mandatory for the final validation phase. This phase involves a controlled pilot study with human volunteers. Strict ethical protocols will be enforced, including obtaining informed consent, practicing data

minimization by collecting only necessary interaction logs, and ensuring participants maintain the right to withdraw at any time.

3.5 Validation Strategy and Performance Metrics The system's efficacy will be evaluated in two stages. First, technical performance will be measured quantitatively using metrics such as extraction accuracy (precision and recall of the LLM schema identifying entities), rendering response latency, and the algorithmic accuracy of the chart suggestion model. Second, the system will undergo a controlled pilot study with volunteer undergraduates. This will provide crucial "Human-in-the-Loop" feedback to evaluate UI usability, the accuracy of the generated content, and the qualitative improvement in active learning and narrative control.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components The proposed solution is an Integrated, Multimodal, and Language-Adaptive Learning Platform designed to address the fragmentation present in current EdTech tools. Rather than isolating content creation, delivery, and assessment, the architecture unifies these processes. The system operates synergistically around a Central Knowledge Orchestrator, which maintains a unified Learner Model and a Knowledge Graph.

The overall system comprises four specialized AI modules:

- **Multimodal Media Factory (Interactive Infographics & Visualizations):** Generates explorable data stories and interactive charts from text.
- **Interactive Audio Experience Tailor:** Transforms transcripts into queryable, bidirectional audio review sessions.
- **Adaptive Assessment Engine:** Utilizes knowledge tracing to generate real-time, multi-format quizzes based on learner proficiency.
- **Multilingual Handwritten Notes Extractor:** Processes English, Tamil, and Sinhala handwritten notes into structured digital formats.

4.2 System Diagram of Components

(Note: In your final document, you will insert the "TAF-Conceptual Diagram.png" here. Below is the textual representation of the architectural flow.)

[Input Layer]

- Source Materials (PDFs, DOCX, PPTX, Plain Text)
- Learner Queries (English, Sinhlish, Thanglish)
- Pedagogical Parameters & Implicit Feedback

[Processing & Orchestration Layer]

- **Data Processing Pipeline:** OCR, Text Parsing, Translation
- **Semantic Engine:** Embedding Generation, Semantic Vectorization
- **Central Storage:** Knowledge Graph Database / Vector Database

[Core AI Modules (Output Generation)]

- Interactive Infographic Generator (Focus Area)
- Audio Synthesis & Q&A
- Adaptive Quiz Generation

[User Interface Layer]

- Personalized Learning Interface (Dashboards, Interactive Canvas)

4.3 Highlighting the Individual Component (Interactive Infographic Generator) The Interactive Infographic Generator is a specialized subsystem designed to combat cognitive overload by turning unstructured text into explorable visual narratives. Its core parts include:

- **Data Extraction Schema:** Utilizes an open-source Large Language Model (LLM) to parse user notes, identifying and extracting key entities, relationships, and statistical data.
- **Chart Recommendation AI:** A lightweight, rule-based AI model that analyzes the extracted data structure to automatically suggest the most suitable visualization template (e.g., hierarchical, timeline, network).
- **Dynamic Visualization Engine:** Built on open-source front-end libraries, this engine renders the templates and embeds interactive capabilities, allowing users to filter data or toggle chart types.
- **Real-Time Collaboration Module:** A synchronization engine enabling multiple users to view, annotate, and discuss the infographic simultaneously.
- **Export Subsystem:** Allows the final visualization to be published as an interactive HTML file or a static image.

4.4 Component Interactions and Data Flow

1. **Ingestion:** A user uploads unstructured study notes or standard textbooks.
2. **Structuring:** The data flows into the NLP pipeline where the open-source LLM extracts the core concepts, mapping them into the central Knowledge Graph.
3. **Visualization Generation:** The Chart Recommendation AI queries the Knowledge Graph to understand the data's dimensionality. It selects a template and passes the structured JSON data to the Visualization Engine.
4. **User Interaction:** The user views the rendered infographic on the Personalized Learning Interface. If the user clicks on a specific node or chart element, the interface sends a query back to the Knowledge Graph to retrieve the exact source quote or deeper context.
5. **Feedback Loop:** User interactions (clicks, time spent on nodes) are recorded as implicit feedback, updating the Learner Model to refine future content complexity.

4.5 Scalability and Security Considerations

- **Scalability:** By utilizing modular, open-source models (such as Gemini 3 Pro/Deepseek via Knowledge Distillation for initial training data generation), the architecture prevents vendor lock-in and allows independent scaling of components. For instance, the LLM processing pipeline can be scaled up on cloud GPUs during heavy document ingestion, while the lightweight Visualization Engine requires minimal server-side rendering, offloading interactions to the client side.
- **Security and Privacy:** To mitigate the privacy and security risks associated with data collection, the system adheres to strict data minimization principles. The platform collects only necessary interaction logs for performance tuning and does not require access to sensitive personal institutional records. Furthermore, processing data through an isolated open-source pipeline reduces the risk of data breaches compared to sending proprietary institutional data to public APIs.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The success of this interactive storytelling and visualization system depends on understanding the needs of its primary users. The core stakeholders for this component include:

- **University Students (Undergraduates):** The primary consumers of the system, specifically those from diverse educational backgrounds or with different learning styles who require guided, self-paced discovery to overcome cognitive overload and rigid curricula.
- **Educators and Instructional Designers:** Users who require intuitive, AI-assisted authoring tools to transform standard text into engaging, explorable visual materials for their students.
- **Lifelong Learners and Researchers:** Individuals engaging with complex research data who need tools to dynamically synthesize and visualize structured relationships.

5.2 Functional Requirements

To achieve the specific objectives of the interactive infographic generator, the system must fulfill the following numbered functional requirements:

1. The system must extract and structure key entities, relationships, and statistics from unstructured user notes or research data using an open-source Large Language Model (LLM).
2. The system shall generate multiple explorable visualization templates, including hierarchical, timeline, and network views, utilizing open-source libraries.
3. The system must provide an interface that allows users to dynamically filter visualized data based on their study preferences.
4. The system shall enable users to click on specific chart elements to instantly drill down into detailed information or retrieve original source quotes.
5. The system must allow users to seamlessly toggle between different visualization types during their active learning session.
6. The system shall incorporate a rule-based or lightweight AI model capable of analyzing the data structure to automatically suggest the most suitable chart type.
7. The system must support real-time collaborative features, allowing multiple users to simultaneously view, annotate, and discuss the same generated infographic.
8. The system shall feature an export module that allows users to publish or download the final visualization as an interactive HTML file or a static image for inclusion in reports.

5.3 Non-Functional Requirements

To ensure the system is effective, secure, and user-friendly, it must adhere to the following non-functional requirements:

- **Performance and Responsiveness:** The system's loading and response times must be minimized. The dynamic filtering and drilling-down interactions within the visual interface must execute without noticeable latency to maintain user engagement.
- **Usability:** The system must feature clear, user-friendly interfaces with minimal distractions, providing proper guidance for students to easily navigate and manipulate complex data visualizations.
- **Security and Privacy:** The system must adhere to strict data minimization principles, collecting only necessary interaction logs for performance tuning and avoiding access to sensitive personal institutional records.
- **Scalability:** The architecture must be scalable to accommodate an increasing number of concurrent users , particularly ensuring that real-time collaborative annotation features remain synchronized and stable under high traffic.
- **Reliability and Availability:** The platform must be highly reliable and available to ensure that students have uninterrupted access to their personalized study materials and visual knowledge bases.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona The primary industry vertical for this proposed system is EdTech, specifically targeting Higher Education and Lifelong Learning. The target market is divided into two core customer personas:

- **The University Undergraduate:** Students, especially those from non-traditional backgrounds or with diverse learning styles, who are disproportionately impacted by rigid, text-heavy curricula and struggle with cognitive overload. They seek guided, self-paced discovery tools.
- **The Educator/Instructional Designer:** University lecturers and teaching staff who require intuitive, AI-assisted course authoring tools to easily translate complex data and standard textbooks into highly engaging, visual formats for their students.

6.2 Value Proposition Current educational tools—including advanced platforms like Google NotebookLM—typically generate static or linear visual summaries from user notes. The core value proposition of our interactive infographic generator is transforming these static charts into dynamic, explorable data stories. We empower users to interrogate the visualizations by filtering data, switching perspectives, and clicking on chart elements to drill down into original source quotes. This level of context-aware interactivity and user-driven narrative control provides a deeply personalized learning experience that passive tools simply cannot match.

6.3 Revenue Model

To successfully commercialize the platform, a Software as a Service (SaaS) revenue model will be adopted, catering to both B2C (Business-to-Consumer) and B2B (Business-to-Business) segments:

- **B2C Subscription (Freemium):** A free tier allowing basic note extraction and standard chart generation to drive initial user adoption among students, followed by a premium monthly subscription unlocking advanced AI template suggestions, unlimited real-time collaboration, and high-resolution export modules.
- **B2B Institutional Licensing:** Enterprise-level annual contracts for universities and educational institutions, providing bulk student access, LMS (Learning Management System) integration, and localized data compliance.

6.4 Cost Estimation Developing this system primarily involves software-based implementation. However, training and hosting the open-source Large Language Models (LLMs) and the visualization rendering engines require significant computational resources. The primary source of cost will be cloud computing membership fees for the processing capacity of virtual machines.

Estimated Budget Allocation Strategy:

- **Cloud Infrastructure & GPU Hosting:** For running the open-source NLP extraction schemas and real-time collaboration engines.
- **Internet Usage and Technical Charges:** For continuous data scraping, model fine-tuning, and API maintenance.
- **Learning Materials & Research Access:** To access premium datasets or academic journals required for algorithm benchmarking.
- **Marketing & User Acquisition:** Advertising the product upon launch to reach the target student demographic.

6.5 Pricing Strategy

The pricing strategy will be competitive and value-based, designed to undercut expensive proprietary tutoring systems while offering superior visual interactivity. The student-focused premium tier will be priced affordably (e.g., comparable to basic streaming services) to ensure accessibility for undergraduates. The institutional tier will employ tiered pricing based on the number of active student licenses, making it a cost-effective solution for university departments seeking to improve retention rates.

6.6 Competitive Advantage The system's primary competitive advantage is its context-aware interactivity combined with a transparent architecture. Unlike Google NotebookLM, which offers linear outputs, this component introduces user-driven narrative control where users actively interrogate the data. Furthermore, the integration of real-time collaborative features allows multiple users to view, annotate, and discuss the same infographic simultaneously. By utilizing open-source models for intelligent structuring and template suggestion, the system creates a transparent, highly customizable alternative to the "black box" proprietary systems currently dominating the market.

6.7 Intellectual Property Considerations Because the system leverages open-source Large Language Models (LLMs) and open-source front-end visualization libraries, the underlying base models cannot be patented. However, intellectual property protection will focus on the unique proprietary elements of the system. This includes obtaining copyrights for the custom-designed data extraction schema, the specific algorithmic logic of the chart recommendation AI, and the unique User Interface/User Experience (UI/UX) source code. Additionally, trademarks will be registered for the platform's brand name and logo to establish a distinct identity within the EdTech market.

7. BUDGET AND JUSTIFICATION

7 BUDGET AND JUSTIFICATION

Developing the interactive infographic generator is primarily a software-based endeavor; therefore, there are no physical hardware components associated with the core implementation. To ensure the successful development and testing of the intelligent data extraction schema and visualization engine, we have planned to utilize cloud computing.

Below is the estimated budget required to complete this specific component of the research project, tailored to the constraints of an undergraduate study.

Table 1: Budget Estimation

Expense Category	Description	Estimated Amount (LKR)
Hardware Costs	Personal workstations and mobile devices for development and testing.	0.00
Software Licenses	Open-source LLMs, Python frameworks, and open-source visualization libraries.	0.00
Cloud Services & APIs	Cloud processing capacity and API token costs for data extraction.	15,000.00
Internet & Connectivity	High-speed data allocation for model training, web scraping, and cloud syncing.	4,000.00
Learning Materials	Access to premium academic journals, research papers, and specific online courses.	5,000.00
Human Effort Cost	Academic research and development hours (Student Project).	0.00
Total Estimated Budget		24,000.00

7.1 Justification for Each Expense

- **Hardware Costs (0.00 LKR):** As this is a software-centric module, all programming, algorithm development, and initial localized testing will be conducted using the existing personal computers of the research group members. No dedicated physical servers or specialized external hardware are required for this phase.
- **Software Licenses (0.00 LKR):** The core novelty of this component relies on its transparent, modifiable architecture. By strictly utilizing open-source tools—such as Python for core programming, Hugging Face for NLP pipelines, and open-source front-end charting libraries—we completely eliminate the need for expensive proprietary software licenses.
- **Cloud Services & APIs (15,000.00 LKR):** This is the primary source of cost for the project. Generating the foundational synthetic dataset requires interacting with advanced models like Gemini 3 Pro and Deepseek via Knowledge Distillation. Furthermore, running the LLM-driven schema to extract key entities and relationships from user notes requires the processing capacity of cloud-based virtual machines, as standard local hardware is insufficient for high-speed, real-time NLP inference.
- **Internet & Connectivity (4,000.00 LKR):** Continuous, high-bandwidth internet access is required to interface with cloud environments, download heavy public benchmarking datasets like "EdNet" and "ASSISTments", and support real-time collaborative feature testing.
- **Learning Materials (5,000.00 LKR):** To gain the essential specialized domain expertise required for this project, funds are allocated for enrolling in specialized online courses (e.g., on Coursera or edX focusing on AI and Learning Analytics) and purchasing access to paywalled industry research reports or specific IEEE/ACM journal papers.
- **Human Effort Cost (0.00 LKR):** Because this system is being developed as a mandatory IT4010 final-year research project, the labor, coding, and testing hours contributed by the undergraduate researchers are strictly considered academic effort and are not financially billed to the project budget.

8. WORK BREAKDOWN STRUCTURE (WBS)

To ensure the successful delivery of the "AI-Powered Personalized Course Creation and Discovery Platform," the project is modularized into four distinct, yet interconnected, AI components. The Work Breakdown Structure (WBS) is designed to allocate an end-to-end development pipeline to each researcher, ensuring equal technical complexity and a balanced workload.

8.1 Task Breakdown Per Member

- **Member 1: Kojithan P.Y (IT22264220) – Interactive Infographic Generator**
 - **Task 1.1:** Design a schema to extract and structure key entities, relationships, and statistics from user notes using an open-source Large Language Model (LLM).
 - **Task 1.2:** Implement multiple visualization templates (hierarchical, timeline, network) using open-source libraries.
 - **Task 1.3:** Develop a dynamic interface allowing users to filter data, click elements to drill down into source quotes, and toggle visualization types.
 - **Task 1.4:** Integrate a rule-based or lightweight AI model to suggest the most suitable chart type.
 - **Task 1.5:** Enable real-time collaborative features for simultaneous viewing and annotation.
 - **Task 1.6:** Create an export module for interactive HTML or static image publishing.

- **Member 2: Baskaran V (IT22172600) – Interactive AI-Powered Audio Review Component**
 - **Task 2.1:** Implement automatic speech recognition (ASR) to transcribe uploaded audio/podcasts.
 - **Task 2.2:** Develop an NLP pipeline using an open-source LLM to analyze transcripts for key concepts, structured summaries, and central themes.
 - **Task 2.3:** Build an interactive web interface with clickable transcript segments for context-specific Q&A.
 - **Task 2.4:** Integrate an open-source Text-to-Speech (TTS) model for verbal AI answers.
 - **Task 2.5:** Conduct system evaluation based on transcription accuracy, response latency, and user engagement.

- **Member 3: Nishara T (IT22223876) – Adaptive Quiz and Assessment Generation Engine**
 - **Task 3.1:** Develop a Knowledge Tracing Model to infer concept mastery from interaction history.
 - **Task 3.2:** Implement an NLP pipeline to auto-generate diverse question types (MCQ, fill-in, short answer).
 - **Task 3.3:** Create an algorithm to dynamically adjust quiz difficulty and focus based on the learner model.
 - **Task 3.4:** Integrate a feedback mechanism offering Socratic hints and scraped web study resources.
 - **Task 3.5:** Evaluate the engine's impact on learning efficacy via pre/post-test scores.

- **Member 4: Sarmitha S (IT22637482) – Multilingual Handwritten Notes Extractor**
 - **Task 4.1:** Collect handwritten notes across English, Tamil, and Sinhala.
 - **Task 4.2:** Preprocess images (denoise, adjust contrast) and prepare labeled training data.
 - **Task 4.3:** Train or fine-tune an AI model for multilingual handwriting recognition.
 - **Task 4.4:** Clean, structure, and convert extracted text into study-ready Canva notes.
 - **Task 4.5:** Store structured outputs in a searchable digital knowledge base.

8.2 Timeline Alignment The project spans the 2026 academic year and is divided into four synchronized phases to ensure all components integrate smoothly into the Central Knowledge Orchestrator:

1. **Phase 1: Planning & Data Collection (Months 1-2):** Literature review, UI/UX design, synthetic data generation, and gathering multilingual note datasets.
2. **Phase 2: Core Model Development (Months 3-5):** Independent development of LLM extraction schemas, knowledge tracing models, and handwriting recognition algorithms.
3. **Phase 3: Integration & UI Implementation (Months 6-7):** Connecting the four standalone AI modules to the unified Personalized Learning Interface dashboard.
4. **Phase 4: Testing & Pilot Study (Months 8-10):** Executing the controlled pilot study with undergraduate volunteers to evaluate performance, followed by ethical clearance reporting and final documentation.

8.3 Balanced Workload Distribution

The workload is equitably distributed by assigning each member a complete, vertically integrated subsystem. Every researcher is responsible for their module's data ingestion (e.g., text, audio, images), underlying AI/ML processing (e.g., NLP extraction, speech recognition, computer vision), user interface integration, and individual performance evaluation. This ensures that all four members engage deeply with core AI domains and software engineering practices without creating technical bottlenecks or unequal dependencies

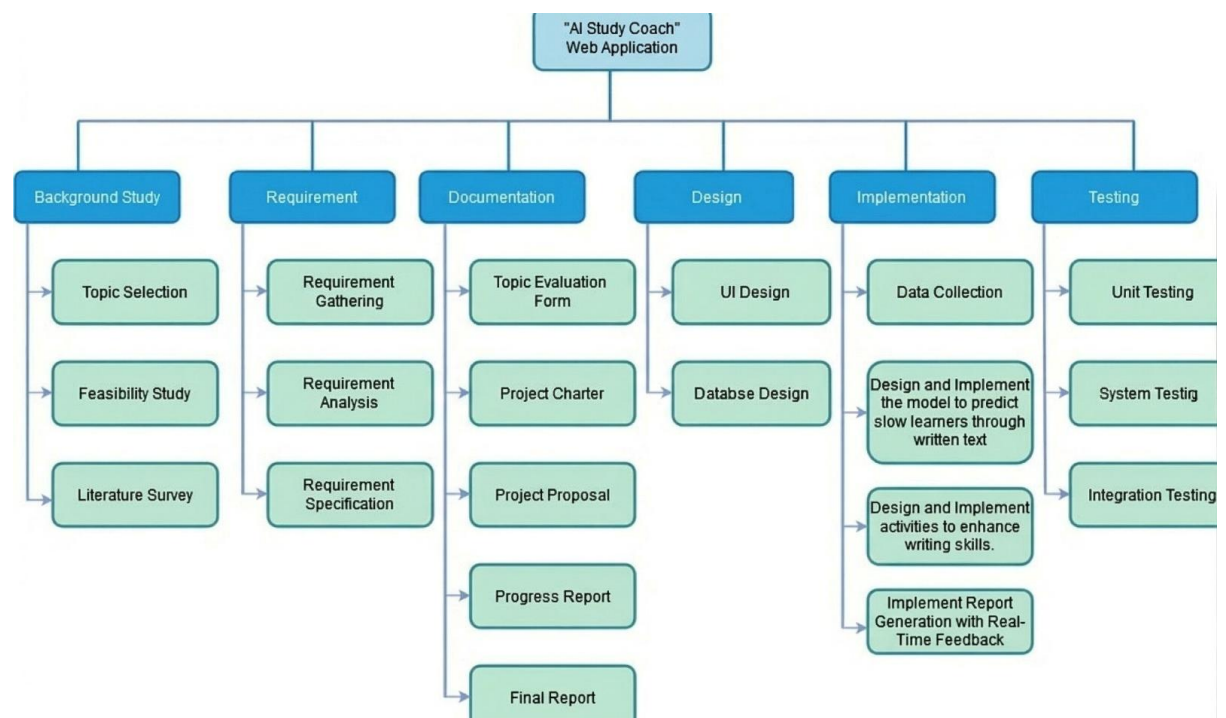


Figure 1 (WBS Chart)

9. GANTT CHART

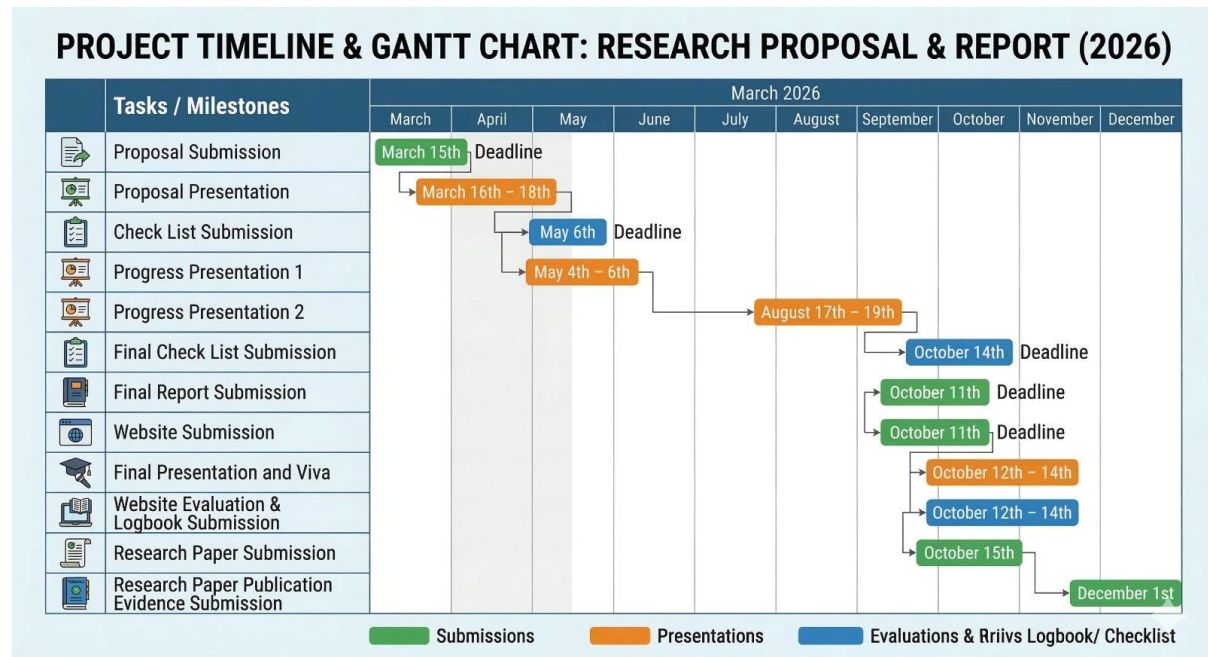


Figure 2: GANTT Chart

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11. APPENDICES

- Survey instruments

1. Online Questionnaires (via Google Forms)

What it is: Digital surveys given to students before and after using your interactive charts.

Why use it: To collect hard numbers and quantitative data (e.g., rating usability on a scale of 1 to 5).

Example: "Rate how easy it was to switch from the Timeline View to the Network View."

2. Structured Interviews

What it is: A one-on-one conversation using a strict, pre-written list of specific questions.

Why use it: To make sure you get qualitative answers on specific features you built (like the AI chart recommendation).

Example: "Did the AI correctly suggest the best chart type for your notes? Please explain why or why not."

3. Unstructured Interviews

What it is: An open, free-flowing conversation with the user, without a strict script.

Why use it: To discover unexpected insights. Sometimes users interact with a visual chart in ways you didn't plan for. This allows the user to freely share their feelings, frustrations, or "aha!" moments.

Example: "Tell me about your experience exploring the data today. What stood out to you?" (And then you just listen and let the conversation flow naturally).

- Detailed diagrams

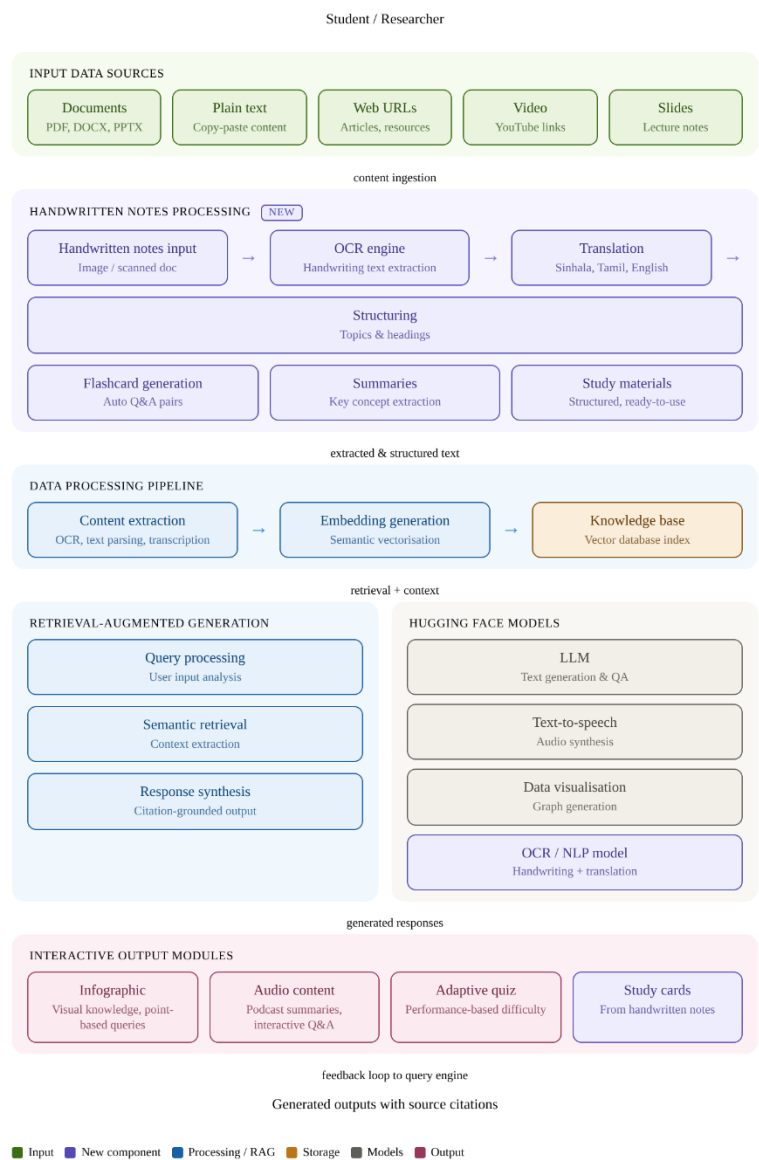


Figure 3: Detailed Diagram

- Preliminary UI mockups

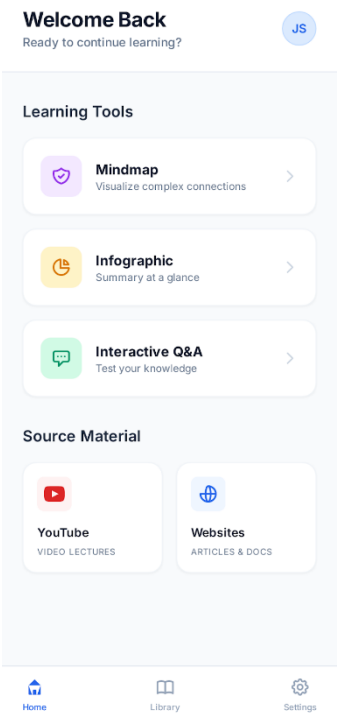


Figure 4: Mockup UI

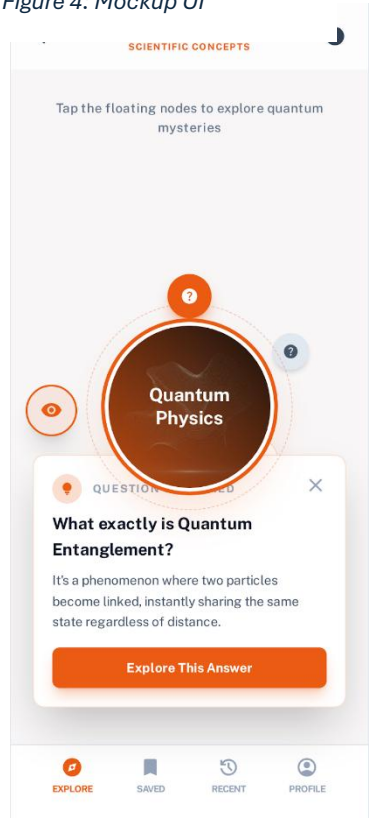


Figure 6: Mockup UI 3

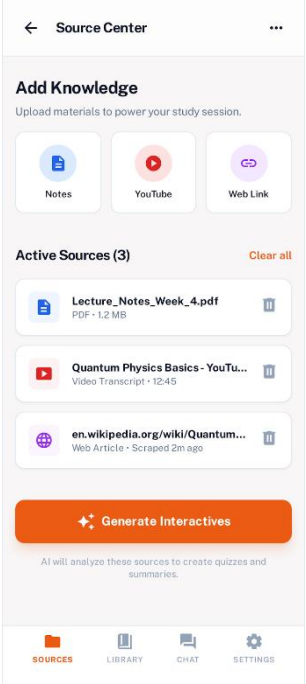


Figure 5: Mockup UI 2

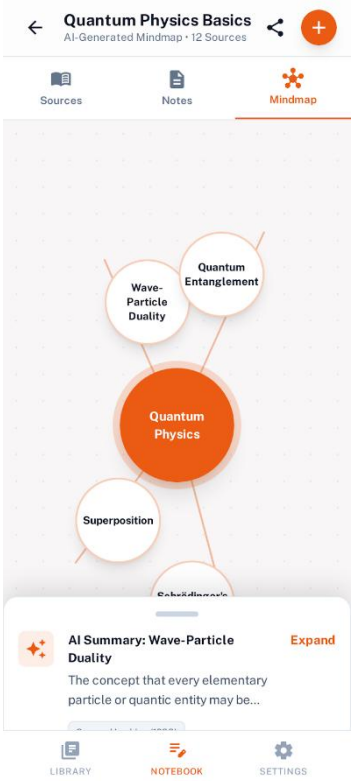


Figure 7: Mockup UI 4

Risk analysis

1. Technical Risk: AI Extraction Errors

- **The Risk:** The open-source Large Language Model (LLM) might misunderstand the student's notes and extract incorrect facts or miss important relationships.
- **The Mitigation:** I will use strict prompt engineering to force the AI to output data in a highly structured format (JSON). If the AI makes a mistake, the user interface will allow the student to easily edit or delete the incorrect data node.

2. Performance Risk: Slow Loading Times (Latency)

- **The Risk:** Rendering complex, interactive charts (like a large network graph) might cause the system to lag or load slowly, which frustrates users.
- **The Mitigation:** I will use lightweight, open-source front-end visualization libraries. The heavy AI processing will be handled on the cloud servers, so the user's computer only has to display the final interactive chart.

3. Scalability Risk: Real-Time Collaboration Crashes

- **The Risk:** When multiple students try to view and annotate the exact same infographic at the same time, the data might fail to sync properly.
- **The Mitigation:** I will use a reliable synchronization engine and thoroughly test the system to ensure it remains stable even when multiple users are interacting with the chart simultaneously.

4. Ethical Risk: Data Privacy Concerns

- **The Risk:** Processing user-uploaded study notes could accidentally collect or expose private or sensitive information.
- **The Mitigation:** I will implement strict "data minimization". The system will only collect basic interaction logs (like where the user clicks to check system performance) and will never access or store sensitive personal institutional records.